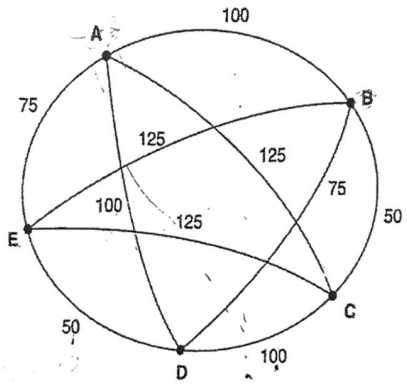
## Question 1 [18]

1.1. Use the truth table to convert the following sentence to propositional logic. (06)

"I will pass Monday's test only if I attend classes or consults after each class."

1.2. When it comes to artificial intelligence (AI), intelligence is still the concept to examine and understand. There are two themes that focuses on intelligence. The first one being the view of intelligence as rooted in culture and society and, consequently, emergent, and the second one being that the intelligence is reflected by the collective behaviours of large numbers of very simple interacting, semiautonomous individuals, or agents. Which themes support an agent-oriented and emergent view of intelligence? (08)

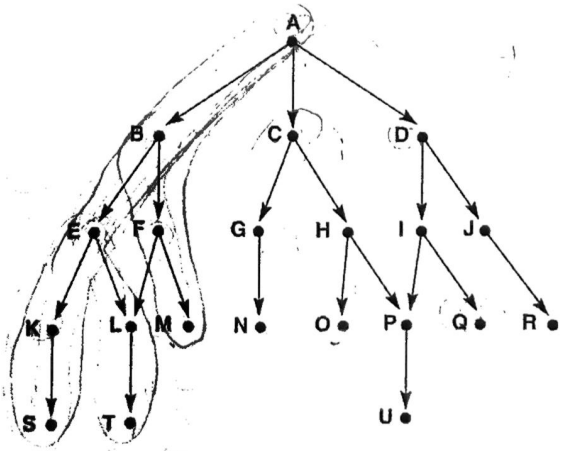
1.3. Use the following diagram to show the nearest neighbour path from A to all the state and show the cost of it. (04)




## Question 2 [22]

2.1. What is the difference between New state list, State List and Dead End in backtracking. (06)

2.2. Use the following tree to find goal Q using backtracking. (08)



2.3. What is the difference of the State space in goal-directed search and data-directed search. (04)

2.4. Use the following sets of propositional calculus expressions to graph. (04)

$P \wedge Q \rightarrow goal$

$R \wedge S \rightarrow P$

$W \wedge R \rightarrow P$

$T \wedge U \rightarrow Q$

$V \rightarrow S$

$Start \rightarrow V \wedge R \wedge Q$